La Scuola d'Italia / Dr. Huson / IB Math 11
20 October 2026

Nickname:

2.2 Classwork: Radicals and Exponents (no calculator)

1. Simplify each expression.

   (a) $3^4 \times 3^2$

   (b) $\dfrac{x^7}{x^3}$

   (c) $(a^2)^3$

   (d) $(2x^3)^2$

   (e) $\dfrac{6x^5}{2x^2}$

   Evaluate each expression.

   (f) 7^0

   (g) 2^{-3}

   (h) $9^{\frac{1}{2}}$

   (i) $8^{\frac{2}{3}}$

   (j) $4^{-\frac{1}{2}}$

2. Simplify each expression. Write each answer in the form $a\sqrt{b}$, where a and b are integers, or as an integer.

   (a) $\sqrt{12}$

   (b) $\sqrt{50}$

   (c) $\sqrt{2} \times \sqrt{8}$

   (d) $3\sqrt{5} + 4\sqrt{5}$

   (e) $\sqrt{18} - \sqrt{2}$

   (f) Rationalize the denominator of $\dfrac{6}{\sqrt{3}}$.

   (g) Expand and simplify $\left(3 - \sqrt{5}\right)\left(3 + \sqrt{5}\right)$.

   (h) Hence, rationalize the denominator of $\dfrac{8}{3 - \sqrt{5}}$.